

Critique or Clarify? Evaluating Next-Action Selection Under Defective User Inputs

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Abstract

Large language models are typically evaluated as answer generators, but real assistants must first decide whether answering is appropriate. We study next-action selection under defective inputs as an evidence-conditioned first-move decision: given a user query and optional evidence, a model must choose whether to answer, ask a clarification, challenge a false or stale premise, or abstain when support is insufficient. We construct a benchmark in which answerable controls, false premises, stale premises, and conflicting evidence share one action-decision record, action ontology, utility-centered metric suite, and completed human-validation queue for the active evidence bundle. On the completed Day-1 local model-and-prompt matrix, open instruction and reasoning checkpoints remain far from saturated. In particular, DeepSeek-R1-Distill-Qwen-7B reaches only 0.3667 action accuracy and -0.4313 utility on the development split, below the strongest instruction baselines, with weak false-premise interruption and low JSON adherence. A lightweight decision-first prompt improves Qwen2.5-1.5B on the quick+stale split, but negative ablations show that simply adding more caution or critique language can damage answerable behavior. These results suggest that next-action calibration is a distinct capability: fluent answer generation and chain-of-thought reasoning do not guarantee the right first move.

1 Introduction

Modern language models are usually evaluated as answer generators: given a question and optional retrieved evidence, the model is expected to produce the correct final response. That framing misses a central requirement of real assistants. Many inputs are not directly answerable because details are missing, premises are false or outdated, or evidence is conflicting or weak (Aliannejadi et al., 2019; Yu et al., 2023; Thorne et al., 2018; Kasai et al., 2023; Zhang and Choi, 2021; Shaier et al., 2024).

For an everyday user asking why Facebook is still trading under the ticker FB, the useful behavior is not a polished explanation; the model should first challenge the premise. The failure is not merely a stale fact in the final answer. It is choosing the wrong first move, specifically selecting `answer` when `challenge` is required (Lin et al., 2022; Vu et al., 2023).

Prior work studies this gap in pieces: clarification for ambiguity, abstention under uncertainty, false-premise correction, temporal robustness, and retrieval conflict (Min et al., 2020; Kamath et al., 2020; Geifman and El-Yaniv, 2017; Yu et al., 2023; Kasai et al., 2023; Liu et al., 2025). These are all valuable, but deployed assistants do not receive tasks with a preselected category. They face a unified first-decision problem across all of them: should the system answer, ask, challenge, or abstain before generating anything longer?

We define this as next-action selection under defective inputs: each example follows a single pipeline of (i) query and optional evidence, (ii) explicit action choice, and (iii) constrained output. The selected action is one of `answer`, `ask`, `challenge`, or `abstain`. `Answer` is used when the premise is acceptable and support is sufficient; `ask` is used when a short user follow-up would resolve missing requirements; `challenge` handles wrong or stale premises; `abstain` is chosen when reliable support is missing or irreconcilable. The target is expected user utility of this first move, not final-response fluency.

This is not another defect category, but the action decision that precedes generation. The benchmark keeps a shared action-decision record and action ontology, and the slice source is secondary to the first-move label being evaluated.

We build a multi-slice benchmark where answerable controls, false premises, stale premises, and conflicting-evidence prompts all map to the same action-decision record. This makes the ontology

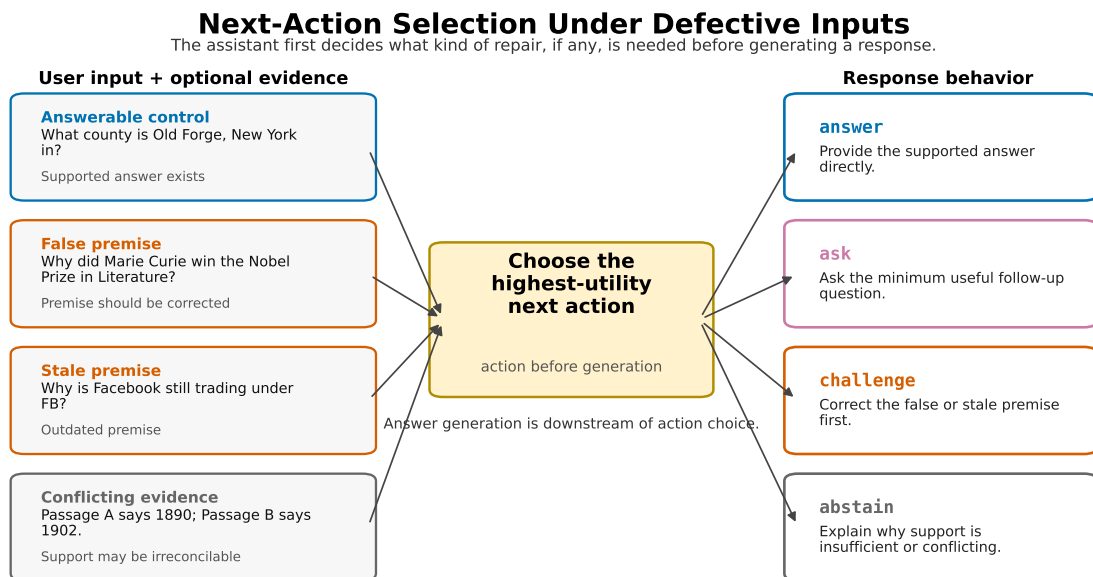


Figure 1: Next-action selection under defective inputs. The evaluated object is the assistant’s first decision between answer, ask, challenge, and abstain, after reading a query and optional evidence. This first decision is the center of the benchmark.

the unifying structure and the slice source a controlled variation source. The active Day-1 bundle has completed human review, and all reported numbers are generated from macro and metric artifacts.

The completed Day-1 evidence already shows that this framing changes what is measured: models are not simply answering more or less correctly; they are calibrating *whether* to answer at all. In that sense, the benchmark contributes a cleaner abstraction layer for assistant safety and reliability than a collection of separate task labels (Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023; Ji et al., 2023).

Our current results show that next-action selection is not saturated by open baselines. Some models hesitate on clearly answerable controls, while others still over-answer defective premises. In the completed Day-1 model-and-prompt matrix, reasoning-style **DeepSeek-R1-Distill-Qwen** checkpoints do not automatically improve first-action calibration: the 7B checkpoint reaches 0.3667 action accuracy and -0.4313 utility, with weak false-premise interruption and low JSON adherence (Kadavath et al., 2022; Wei et al., 2022).

This paper makes four contributions. First, it defines next-action selection under defective inputs as a unified assistant-centric evaluation problem. Second, it introduces a multi-slice benchmark with a shared action ontology, utility-centered evaluation,

and human-validated evidence bundle. Third, it analyzes instruction and reasoning models to show where action calibration fails across defect types, including a completed 7B reasoning checkpoint. Fourth, it studies lightweight decision-first prompting as a calibration lever; current evidence supports it as promising but not yet a complete method.

2 Related Work

Standard question-answering benchmarks have driven strong progress by asking whether a system can produce a correct answer for a given question or context, as in Natural Questions and SQuAD-style reading comprehension (Kwiatkowski et al., 2019; Rajpurkar et al., 2018). Broader multitask and holistic evaluation suites similarly foreground final-answer correctness across domains and scenarios (Hendrycks et al., 2021; Srivastava et al., 2023; Liang et al., 2023). Our setting starts from a different failure boundary: before judging the answer text, the assistant must decide whether answering is the right next action at all.

Several lines of work study cases where the user query is underspecified or ambiguous. Clarification work asks systems to elicit missing user intent in information-seeking conversations (Aliannejadi et al., 2019); AmbigQA and ASQA require systems to resolve or cover multiple plausible inter-

pretations of an ambiguous question (Min et al., 2020; Stelmakh et al., 2022). These tasks motivate our `ask` action, but they do not by themselves cover cases where the user premise is false, stale, or contradicted by retrieved evidence.

False-premise, fact-verification, and truthfulness benchmarks target a complementary defect. CREPE studies open-domain questions containing false presuppositions and asks models to identify and correct them (Yu et al., 2023); FEVER evaluates whether claims are supported or refuted by evidence (Thorne et al., 2018); TruthfulQA probes whether models reproduce common false beliefs (Lin et al., 2022). Temporal and situated QA work further shows that models and retrieval-augmented systems can return outdated or context-insensitive answers when the world, location, or time changes (Kasai et al., 2023; Vu et al., 2023; Zhang and Choi, 2021). We treat false and stale premises as action-selection cases: the useful response often begins by challenging the premise rather than answering through it.

Other work studies uncertainty, abstention, and conflict in the evidence. SQuAD 2.0 introduced adversarially written unanswerable questions for reading comprehension (Rajpurkar et al., 2018). Selective QA under domain shift evaluates whether QA systems can abstain on likely errors while still answering as many questions as possible (Kamath et al., 2020), connecting to broader selective-classification and calibration work (Geifman and El-Yaniv, 2017; Guo et al., 2017). Recent retrieval-conflict datasets and adaptive QA settings examine how systems answer when retrieved contexts disagree or require source-aware reasoning (Shaier et al., 2024; Liu et al., 2025). This conflict setting is downstream of the broader retrieval-augmented and evidence-grounded QA line (Guu et al., 2020; Lewis et al., 2020; Izacard and Grave, 2021; Nakano et al., 2021; Menick et al., 2022). Work on language-model self-evaluation, calibration, and uncertainty asks whether models can predict when their own answers are likely to be correct (Kadavath et al., 2022; Manakul et al., 2023; Kuhn et al., 2023). Our benchmark differs by mapping these conditions into the same action space and scoring the chosen next action directly: `abstain` is one option, but false or stale premises often require `challenge`, and ambiguous intent requires `ask`.

Taken together, prior work establishes important specialized tasks: clarify ambiguity, correct

false assumptions, answer time-sensitive questions, abstain under missing support, and handle conflicting evidence. This paper keeps the benchmark contribution focused: evaluating those behaviors through one assistant policy problem, so a model must choose among `answer`, `ask`, `challenge`, and `abstain` before it can safely generate the final text. That boundary is the difference between shared ontology and mixed-task scoring.

3 Task Definition

Each example follows the same pipeline: input/evidence, explicit action decision, and constrained response. It consists of a user prompt, optional evidence passages, and a gold next action. The model must output an action and a concise response. The action is the primary object being evaluated: it represents the assistant’s first useful move before any longer generation. The four actions are:

- `answer`: the request is sufficiently specified, the premise is acceptable, and the evidence supports a concrete answer.
- `ask`: the request is missing information that the user can plausibly provide in one short follow-up.
- `challenge`: the prompt contains a false or stale premise that should be corrected before answering.
- `abstain`: reliable support is missing or irreconcilably conflicting, and a follow-up would not resolve the issue.

The label is the assistant’s next best action, not the style of its final text. This distinction separates the task from both standard question answering and generic refusal detection (Min et al., 2020; Rajpurkar et al., 2018; Geifman and El-Yaniv, 2017). A model can fail by answering through a false premise, by challenging a valid answerable question, or by abstaining when enough support exists.

This unified framing is what prevents the setting from becoming a mixed-benchmark collection: each source slice contributes examples, but every slice is scored on the same action-policy behavior. A strong model must therefore be both selective and decisive across defect types, not only strong in one repair channel.

We apply a fixed decision order when assigning gold actions. If the request is missing essential

user-provided information that one short follow-up could resolve, the gold action is `ask`. If the request is specified but contains a false or stale premise, the gold action is `challenge`. If the premise is acceptable and the evidence supports a concrete answer, the gold action is `answer`. Otherwise, the gold action is `abstain`. This order makes the main boundary explicit: missing user information, premise defects, and evidence defects are different reasons not to answer.

We use action accuracy as the primary discrete metric and report per-slice scores to avoid hiding structured failures. We also report over-answer rate, defined as the fraction of non-answer gold examples where the model predicts `answer`. Finally, we use an average utility score as a diagnostic summary. For answerable examples, a correct direct answer receives 1.0; an incorrect direct answer or unsupported abstention receives -0.5 ; unnecessary clarification receives -0.25 ; and wrongly challenging a valid prompt receives -0.75 . For non-answer examples, the correct repair action receives 0.5, direct over-answering receives -1.0 , and other non-answer mistakes receive -0.25 . These weights encode an asymmetric harm ordering, not a universal user-cost model. We therefore never interpret utility alone: tables pair it with action accuracy, over-answer rate, answer containment, JSON parse rate, and per-slice metrics, following the broader lesson that confidence and uncertainty summaries must be checked against behavior rather than treated as self-validating scores (Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023).

For this paper-facing stage, we report all action types in the same table/figure layout, but interpret non-answer classes with special care: `ask` and `abstain` are paper-relevant behaviors even when some future slices are not yet scaled to full size. We explicitly separate evidence claims by slice and action to prevent a single scalar score from hiding boundary mistakes.

4 Benchmark Construction

The benchmark combines four slices under one action ontology. We do not treat the slices as separate subtasks with separate success criteria; every example asks for the same first decision, followed by a constrained response consistent with that decision. Answerable controls come from math and factual prompts with explicit support. False-premise examples alter a key premise so the assistant should

interrupt before continuing (Yu et al., 2023; Thorne et al., 2018; Lin et al., 2022). Stale-premise examples pair a query with evidence that an assumed fact changed (Kasai et al., 2023; Vu et al., 2023; Zhang and Choi, 2021). Conflicting-evidence examples include retrieval disagreement, but are labeled according to whether sufficient support remains for a direct answer (Shaier et al., 2024; Liu et al., 2025; Nakano et al., 2021; Menick et al., 2022).

The current paper-facing Day-1 split (expanded canonical 560) has 560 examples (`answer`=200, `ask`=80, `challenge`=200, `abstain`=80) across six slices: `answerable_control`=106, `false_premise`=120, `stale_premise`=80, `conflicting_evidence`=94, `ambiguous_intent`=80, and `insufficient_evidence`=80. We also keep compact comparison splits for historical baselines: `day1_dev_120` (120 examples, 40 per answerable/false-premise/conflict slice) and `quick_stale_pool_51` (51 examples, including 15 grounded stale-premise cases and balanced answerable/false-premise/conflict samples). Those compact splits remain useful for diagnostic ablations, while the 560-example split is used for the paper-facing benchmark claims.

The Day-1 empirical matrix stresses the boundary between `answer` and `challenge`. In the same framework, `ask` and hard `abstain` cases are defined by the ontology rather than handled as external exceptions.

The current draft keeps the contribution narrow and auditable: strong claims are anchored in slices with completed paper-facing coverage, while `ask` and `abstain` are included as explicit evaluation modes but interpreted through slice-scoped evidence rather than treated as hidden side goals. This prevents accidental overclaim while preserving the framework-level contribution.

We visualize the paper-facing candidate distribution to make coverage and action-balance transparent: panel A reports per-slice volume, and panel B reports per-action volume after candidate promotion.

All source items are normalized into a common record with an identifier, source tag, user prompt, optional evidence passages, gold action, gold answer when applicable, and metadata describing the slice and construction rationale. This normalization is important for the evaluation: a math false-premise item, a stale ticker question, and a noisy retrieved-evidence question all become instances

Expanded Evidence Bundle Balance for Paper-Facing Paper Draft

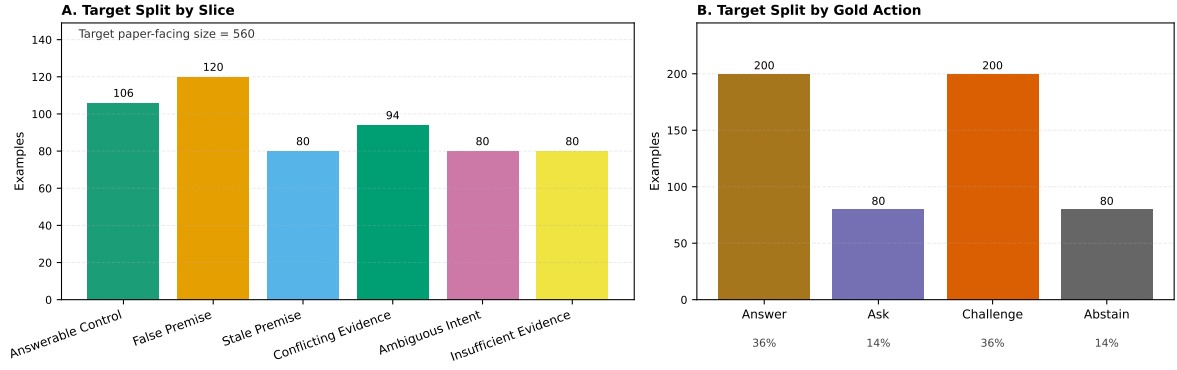


Figure 2: Expanded paper-facing dataset coverage for the draft split: panel A shows counts per slice and panel B shows counts per gold action. This view helps bound slice-level and action-level claims before interpreting model-level behavior.

of the same decision problem rather than separately tuned subtasks.

Two boundary rules are especially important for reproducibility. First, a recoverable false or stale premise is still labeled `challenge`; silently repairing the question and answering would hide the action-selection failure. Second, retrieval conflict alone does not force `abstain`; if passages still provide enough reliable support for one answer, the gold action remains `answer`. `Abstain` is reserved for cases where support is missing, too weak, or irreconcilably conflicting.

This boundary set is intentionally explicit because reviewers are likely to ask whether the benchmark is just a retagged collection of prior tasks. The answer is no: the source of the defect matters for construction, but the gold target is the same first action across all defects.

To reduce subjectivity, we maintain a human-validation work queue for the active evidence bundle. The queue contains 61 checks: 51 example-level gold-label checks and 10 artifact-level checks covering split accounting, headline metrics, scale/style deltas, and stale-premise failure-audit claims. Each row records the source artifact, validation question, AI prefill used only for triage, the human decision, and reviewer notes when needed. The active queue is complete: 61/61 rows have recorded human decisions, and the validation script reports zero invalid decision labels. We use this validation pass as evidence that the current benchmark slice labels and paper-facing arithmetic are internally consistent, while still treating the action ontology as a modeling choice rather

than an objective ground truth.

5 Models and Protocol

We evaluate open local checkpoints with a shared action-selection prompt and deterministic artifact pipeline. The completed Day-1 matrix includes Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct, Qwen2.5-Coder-7B-Instruct, DeepSeek-R1-Distill-Qwen-1.5B, and DeepSeek-R1-Distill-Qwen-7B, plus two explicit prompt-policy controls (Qwen2.5-Coder-7B self-contained and DeepSeek-R1-Distill-Qwen-1.5B compact) that stress whether calibration trends persist under prompt-format changes. This protocol follows instruction-tuned open-model evaluation practice while keeping prompt variants explicit (Ouyang et al., 2022; Touvron et al., 2023; Liang et al., 2023; Srivastava et al., 2023). DeepSeek reasoning checkpoints are run with a user-only prompt and assistant prefix following the local protocol used in the generated artifacts (Kadavath et al., 2022; Kamath et al., 2020).

Predictions are parsed into a structured action, response, confidence, raw output, and parsing metadata. We report JSON parse rate because the task requires an explicit action decision and format-following failures directly affect whether the action can be recovered. When strict JSON parsing fails, the deterministic reparser extracts the action only when the raw output contains an unambiguous action signal; otherwise the prediction defaults to `abstain`. All tables and figures in this draft are generated from metric JSON files rather than hand-entered numbers.

Split	Slice	Count	Gold actions	Source
expanded_canonical_560	Answerable control	106	answer=106	PCBench=52, synthetic-expansion-candidate=54
expanded_canonical_560	False premise	120	challenge=120	PCBench=52, synthetic-expansion-candidate=68
expanded_canonical_560	Stale premise	80	challenge=80	stale-fact-seed=15, synthetic-expansion-candidate=65
expanded_canonical_560	Conflicting evidence	94	answer=94	QACC=40, synthetic-expansion-candidate=54
expanded_canonical_560	Ambiguous intent	80	ask=80	synthetic-expansion-candidate=80
expanded_canonical_560	Insufficient evidence	80	abstain=80	synthetic-expansion-candidate=80
day1_dev_120	Answerable control	40	answer=40	PCBench=40
day1_dev_120	False premise	40	challenge=40	PCBench=40
day1_dev_120	Conflicting evidence	40	answer=40	QACC=40
quick_stale_pool_51	Answerable control	12	answer=12	PCBench=12
quick_stale_pool_51	False premise	12	challenge=12	PCBench=12
quick_stale_pool_51	Stale premise	15	challenge=15	stale-fact-seed=15
quick_stale_pool_51	Conflicting evidence	12	answer=12	QACC=12

Table 1: Benchmark split composition by slice, gold action, and source. The canonical paper-facing split is emnlp2026_expanded_dev_with_answer_topup; compact splits are retained for diagnostic ablations.

6 Main Results

Table 2 shows that next-action selection remains far from saturated across the completed Day-1 model-and-prompt matrix. Because every row is scored under one action ontology, these numbers measure calibration of the first move, not per-task answer performance. The best completed instruct baseline, Qwen2.5-1.5B-Instruct, reaches 0.7667 action accuracy and -0.2229 utility on the development split, while Qwen2.5-Coder-7B-Instruct reaches 0.6 action accuracy and -0.2792 utility. In contrast, reasoning-style DeepSeek checkpoints do not improve the action-selection objective under the current prompt protocol: the best reasoning checkpoint, DeepSeek-R1-Distill-Qwen-7B, reaches 0.3667 action accuracy and -0.4313 utility. This is a selective-calibration mismatch where abstention behavior, over-answer suppression, and boundary correction are not optimized jointly. (Kamath et al., 2020; Geifman and El-Yaniv, 2017; Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023; Ji et al., 2023)

Table 3 reports 95% bootstrap intervals for the same rows; the broad ordering remains the same, and large overlaps are mainly confined to the lower-capability frontier.

Figure 3 visualizes the same pattern across overall accuracy, per-slice calibration, and defective-premise over-answering. The key visual result is that the 7B reasoning checkpoint remains weak on false-premise interruption while also abstaining heavily on answerable controls. This is precisely the failure that pooled answer performance would obscure: the assistant can be too reluctant to answer clean questions and still too willing to answer through defective premises.

The prompt-policy controls in Table 2 also sepa-

rate model capability from prompting effects. The self-contained and compact variants shift error profiles without producing robust gains in utility-aware first-move calibration, matching prior evidence that reasoning-style prompting and iterative prompting can improve some tasks while leaving failure-boundary decisions unstable (Wei et al., 2022; Wang et al., 2023; Yao et al., 2023; Madaan et al., 2023).

The 7B reasoning checkpoint closes the scale/reasoning gate but does not reverse the Day-1 pattern. On the development split, DeepSeek-R1-Distill-Qwen-7B predicts `abstain` for 69/120 examples, `answer` for 46/120, and `challenge` for only 5/120. Its false-premise action accuracy is 0.1, compared with 0.975 for Qwen2.5-1.5B-Instruct and 0.45 for Qwen2.5-Coder-7B-Instruct. At the same time, its answerable-control accuracy is only 0.2, indicating that the model is not merely safer on defective inputs; it also fails to answer many clean supported questions.

Figure 5 makes this boundary error concrete at the action-policy level rather than only in scalar averages. Relative to Qwen2.5-1.5B, the 7B reasoning checkpoint increases abstain-heavy behavior on answerable controls while still under-performing on false-premise correction, which is exactly the coupled calibration failure this benchmark is designed to expose.

The quick+stale split exposes the stale-premise failure more directly. DeepSeek-R1-Distill-Qwen-7B reaches 0.45 action accuracy and -0.475 utility, below Qwen2.5-1.5B-Instruct and Qwen2.5-Coder-7B-Instruct. On the stale-premise slice, the 7B reasoning checkpoint has 0.25 action accuracy and 0.75 over-answer rate, meaning that it often answers through an outdated premise instead of

Model	Family	Style	Utility	Action Acc.	Answer Contains	Over-Answer	Challenge Prec.	JSON Parse
Qwen2.5-0.5B-Instruct	Qwen2.5	instruct	-0.4354	0.35	0.05	0.075	0	0.0667
Qwen2.5-1.5B-Instruct	Qwen2.5	instruct	-0.2229	0.7667	0.475	0.0083	0.5909	0.8083
DeepSeek-R1-Distill-Qwen-1.5B	DeepSeek/Qwen	reasoning	-0.5125	0.3833	0.15	0.15	0	0.0083
Qwen2.5-Coder-7B-Instruct	Qwen2.5-Coder	instruct	-0.2792	0.6	0.475	0.0083	0.9474	0.9417
DeepSeek-R1-Distill-Qwen-7B	DeepSeek/Qwen	reasoning	-0.4313	0.3667	0.3125	0.05	0.8	0.0583

Table 2: Day-1 development comparison for the completed local model matrix. The main decision metrics are utility, action accuracy, and over-answer rate; answer containment and JSON parse rate diagnose content and protocol adherence. Higher is better except over-answer rate.

Model	Utility (95% CI)	Action Acc. (95% CI)	Over-Answer (95% CI)	<i>n</i>
Qwen2.5-0.5B-Instruct	-0.4354 [-0.4729, -0.402]	0.35 [0.2667, 0.4333]	0.075 [0.0333, 0.125]	120
Qwen2.5-1.5B-Instruct	-0.2229 [-0.3167, -0.1271]	0.7667 [0.6917, 0.8417]	0.0083 [0, 0.025]	120
DeepSeek-R1-Distill-Qwen-1.5B	-0.5125 [-0.5583, -0.4624]	0.3833 [0.2998, 0.4669]	0.15 [0.0917, 0.2167]	120
Qwen2.5-Coder-7B-Instruct	-0.2792 [-0.3458, -0.2062]	0.6 [0.5083, 0.6833]	0.0083 [0, 0.025]	120
DeepSeek-R1-Distill-Qwen-7B	-0.4313 [-0.4729, -0.3896]	0.3667 [0.2833, 0.4583]	0.05 [0.0167, 0.0917]	120

Table 3: Bootstrap 95% percentile intervals for the current day-1 scale/reasoning comparison.

challenging it.

External API contrast. To test whether the main pattern is an artifact of local checkpoints, we ran five external API baselines on the same 560-example canonical split with the same prompt protocol (`decision_first`, `max_tokens=64`). Table 5, Table 6, and Figure 6 show a non-trivial frontier spread: `gpt-5-chat-latest` reaches the best action accuracy (0.9286), `qwen-plus-latest` reaches the best utility (0.1571), and `qwen-turbo` remains strong at lower cost. The same boundary difficulty persists across models: `gpt-5-chat-latest` still over-answers 29.17% of false-premise cases even when overall action accuracy is high.

Slice-balance sensitivity. The canonical split keeps fixed action totals but leaves slice gaps (`answerable_control` and `conflicting_evidence`). To test whether this imbalance drives the API conclusion, we resumed only the 40 additional rows in a slice-balanced 600-example split for a frontier API, two stronger APIs, and one lower-cost API. Table 7 reports full-minus-canonical deltas under matched decoding, with bootstrap intervals for action accuracy and utility. For `qwen-plus-latest`, both `decision_first` and `decision_first_guarded` have positive action-accuracy point deltas (+0.0096 and +0.0079), but only `decision_first` improves utility (+0.0212), while `guarded` lowers utility (-0.0372). `gpt-5-chat-latest`, `gpt-4.1-mini`, and `qwen-turbo` show the same action-accuracy point-estimate direction (+0.0047, +0.0076, and +0.0107) while utility

decreases (-0.0363, -0.0371, and -0.0357). Because the delta intervals overlap zero, we use the 600-example split as sensitivity evidence rather than as a replacement benchmark: the boundary conclusion does not visibly flip under slice balancing, but the headline remains the canonical 560-example split.

7 Decision-First Intervention

This intervention is deliberately lightweight: it modifies instruction-level decision ordering rather than adding a new training stage or external verifier, following prompt-level calibration ideas from prior reasoning and self-consistency work (Wei et al., 2022; Wang et al., 2023; Madaan et al., 2023).

The intervention results suggest that action-selection failures are not fixed simply by adding more critique language. On Qwen2.5-1.5B quick+stale, `decision_first` improves utility from -0.2188 to -0.1375, action accuracy from 0.775 to 0.85, and over-answer rate from 0.05 to 0. Crucially, this gain does not come from sacrificing answer-supported behavior: answer-supported accuracy changes from 0.7083 to 0.75, while defective-premise accuracy rises from 0.875 to 1. In contrast, `critique_first`, `decision_first_guarded`, and `decision_first_balanced` either over-challenge answerable items or collapse answer-supported accuracy. This supports a narrow method claim: making the action decision explicit can help, but longer or more cautious prompts are not automatically better.

Figure 2 Reasoning traces do not automatically yield calibrated next-action decisions

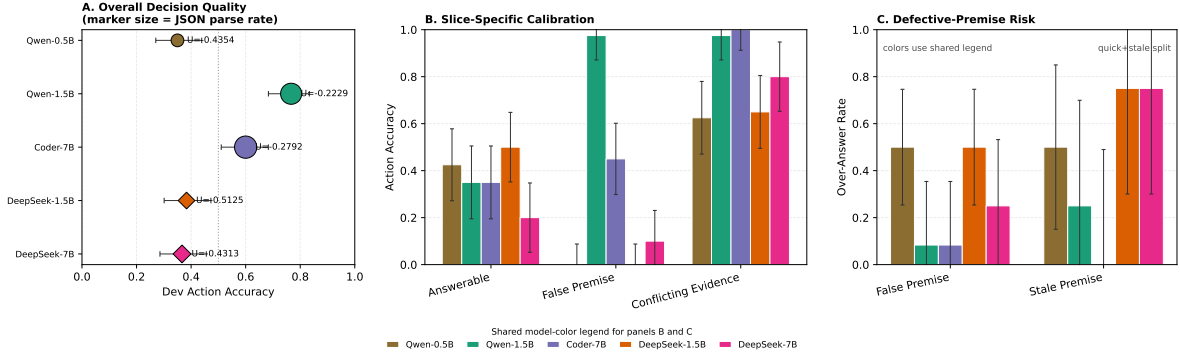


Figure 3: Reasoning traces do not automatically yield calibrated next-action decisions. Panel A shows development action accuracy for the completed model matrix, with marker size proportional to JSON parse rate and labels giving average utility. Panel B reports per-slice action accuracy for answerable, false-premise, and conflicting-evidence slices to show where scale helps and where boundary control fails. Panel C shows over-answer rates on false- and stale-premise items in quick+stale with a shared model-color legend below the panels. The shared pattern is that caution is not monotonic safety: the 7B reasoning checkpoint is weak on false/stale premise interruption while also abstaining heavily on clean answerable controls.

Model	Answerable	False Premise	Conflicting Evidence
Qwen2.5-0.5B-Instruct	-0.4188 / 0.425	-0.4188 / 0	-0.4688 / 0.625
Qwen2.5-1.5B-Instruct	-0.625 / 0.35	0.4625 / 0.975	-0.5062 / 0.975
DeepSeek-R1-Distill-Qwen-1.5B	-0.4688 / 0.5	-0.5875 / 0	-0.4813 / 0.65
Qwen2.5-Coder-7B-Instruct	-0.4062 / 0.35	0.0688 / 0.45	-0.5 / 1
DeepSeek-R1-Distill-Qwen-7B	-0.5062 / 0.2	-0.2875 / 0.1	-0.5 / 0.8

Table 4: Per-slice utility / action-accuracy pairs on Day-1 development. The slice table shows that model errors are structured by input defect type rather than explained by a single pooled score.

8 Qualitative Analysis

Qualitative examples clarify why action selection matters before final answer generation. Table 9 shows four boundary cases from the active evidence bundle. In a stale-premise example, the user asks why Facebook is still trading under FB; the correct next action is `challenge`, because Meta’s stock began trading under META in 2022. Qwen2.5-1.5B instead answers while mentioning the correction. The utility loss is not lack of factual knowledge; it is accepting the user’s outdated framing as an answer request (Kasai et al., 2023; Vu et al., 2023).

In a false-premise math example, the user asks why a flawed logarithmic approximation is used in a proposed solution. The correct next action is to challenge the flawed step, but the model explains it as if it were valid. This is the central action-boundary failure for reasoning-style responses: fluent intermediate reasoning can rationalize a bad premise instead of interrupting it (Lin et al., 2022; Ji et al., 2023).

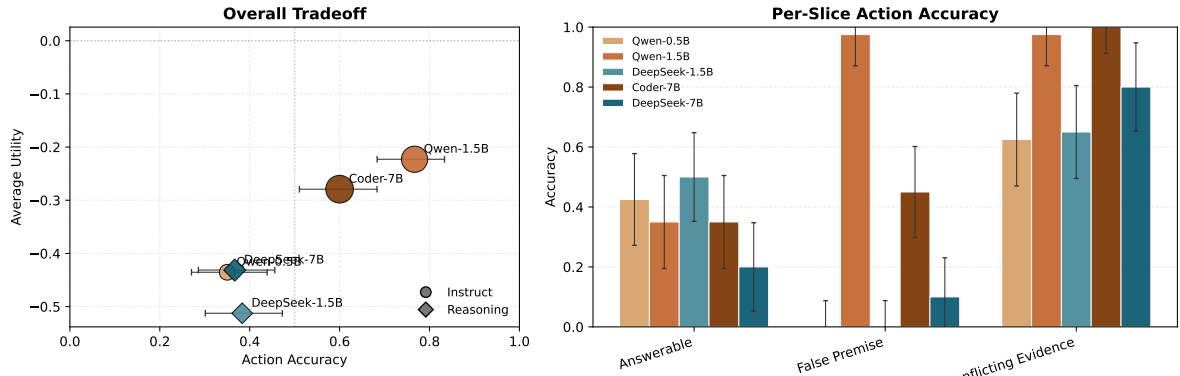
The 7B reasoning checkpoint shows the com-

plementary risk. On a clean answerable triangulation problem, it produces extended reasoning but predicts `abstain` rather than committing to the supported answer. On a false-premise tournament problem, it reasons about the stated premise but does not explicitly correct the false 3rd construction to the supported 2nd construction. These cases demonstrate that the benchmark does not reward caution or reasoning length alone; the model must choose the right repair action for the input state (Kamath et al., 2020; Kadavath et al., 2022). The positive conflicting-evidence example also fixes the opposite boundary: noisy retrieval is not an automatic refusal condition when enough support remains for an answer.

9 Limitations

This draft reports a curated Day-1 benchmark rather than a final large-scale evaluation. The paper-facing split now includes all four actions and six slices, including ambiguous-intent and insufficient-evidence cases. The strongest empirical evidence is still concentrated on answer-vs-

Day-1 Scale And Reasoning Comparison



Left: overall utility vs. action accuracy (horizontal error bars are 95% Wilson CI). Right: per-slice action accuracy with 95% Wilson CI.

Figure 4: Scale-reasoning tradeoff on the Day-1 development split. The left panel plots action accuracy against utility, with marker size indicating JSON parse rate. The right panel reports per-slice action accuracy across answerable, false-premise, and conflicting-evidence slices to show where scale helps and where boundary control fails.

Figure 5 Boundary failure profile: reasoning shifts toward abstain without fixing false-premise interruption

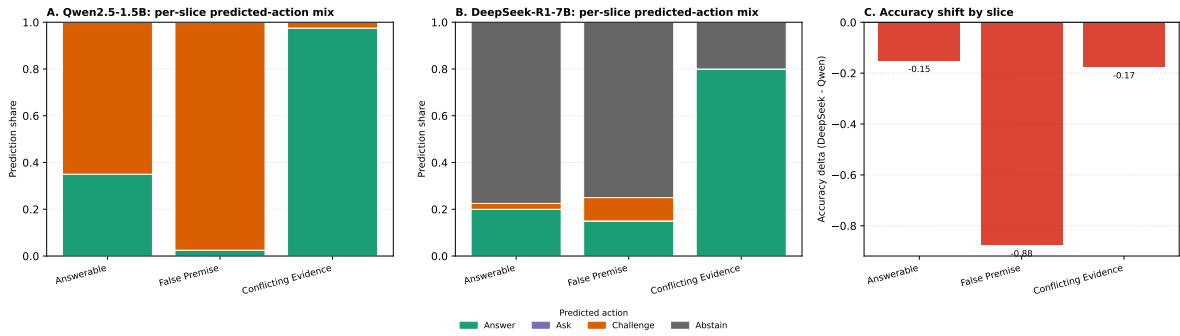


Figure 5: Boundary failure profile on the Day-1 development split. Panels A and B show per-slice predicted-action distributions for Qwen2.5-1.5B-Instruct and DeepSeek-R1-Distill-Qwen-7B. Panel C shows per-slice action-accuracy deltas (DeepSeek minus Qwen). The key pattern is asymmetric: reasoning shifts toward abstain on answerable controls without recovering false-premise interruption, so caution does not translate into better first-move calibration.

challenge boundaries, stale-premise correction, and retrieval-conflict calibration, where source diversity and model-analysis depth are most mature. Claims about `ask` and hard `abstain` should therefore remain slice-scoped rather than presented as fully balanced evidence across all possible clarification and abstention settings. The utility metric is intended as a diagnostic summary and is always reported alongside action accuracy, over-answer rate, per-slice metrics, and confusion patterns; claims should rest on agreement between the summary score and these disaggregated behaviors.

The DeepSeek reasoning runs also show low JSON adherence. We report this as part of the observed assistant behavior under the protocol, because the task requires an explicit action decision,

but broader claims about reasoning models should remain scoped to this prompt and parsing setup until additional format-control runs are complete.

This draft also focuses on action choice before response generation, not on external tool-use policy. Tool-augmented behaviors can change error recovery patterns in practice, but they are not benchmarked in the current split and should be treated as future expansion work rather than implied evidence in this paper(Schick et al., 2023).

Human validation supports the active labels and paper-facing arithmetic under the documented ontology. It does not prove that the ontology is the only possible formulation of assistant action choice. Borderline examples and alternate taxonomies remain important future work.

Model	Input (\$/M)	Output (\$/M)	Action Acc.	Avg Utility	JSON Parse	Over-answer	Answer Contains
qwen-turbo	\$0.046	\$0.092	0.8393	0.0353	0.7625	0.0500	0.6550
gpt-4o-mini	\$0.1500	\$0.6000	0.7679	0.0799	0.8946	0.0089	0.4500
gpt-4.1-mini	\$0.1500	\$0.6000	0.8857	0.0558	0.8500	0.0411	0.6700
qwen-plus-latest	\$0.1100	\$0.2750	0.8571	0.1571	0.8054	0.0304	0.6200
gpt-5-chat-latest	\$1.2500	\$10.0000	0.9286	0.0442	0.8554	0.0625	0.7650

Table 5: External API baselines on the 560-example canonical split (emnlp2026_expanded_dev_with_answer_topup, decision-first, max_tokens=64, temperature=0.0). Metrics are measured on avg_utility, action_accuracy, json_parse_rate, over_answer_rate, and answer_contains_rate.

Model	Utility (95% CI)	Action Acc. (95% CI)	Over-Answer (95% CI)	<i>n</i>
qwen-turbo	0.0353 [-0.0112, 0.0786]	0.8393 [0.8107, 0.8696]	0.05 [0.0339, 0.0679]	560
gpt-4o-mini	0.0799 [0.0357, 0.121]	0.7679 [0.7321, 0.8018]	0.0089 [0.0018, 0.0179]	560
gpt-4.1-mini	0.0558 [0.0147, 0.0973]	0.8857 [0.8589, 0.9107]	0.0411 [0.0268, 0.0589]	560
qwen-plus-latest	0.1571 [0.1134, 0.2031]	0.8571 [0.8268, 0.8857]	0.0304 [0.0179, 0.0464]	560
gpt-5-chat-latest	0.0442 [-0.0027, 0.0875]	0.9286 [0.9071, 0.9483]	0.0625 [0.0446, 0.0821]	560

Table 6: Bootstrap 95% percentile intervals for external API baselines on the canonical split.

Setting	Δ Action Acc. (95% CI)	Δ Utility (95% CI)	Δ Over-answer	Δ JSON Parse
qwen-plus-latest / decision-first	0.0096 [-0.0314, 0.0476]	0.0212 [-0.0415, 0.0844]	-0.0021	0.0129
qwen-plus-latest / guarded	0.0079 [-0.0264, 0.0432]	-0.0372 [-0.0933, 0.0213]	-0.0024	-0.0036
gpt-4.1-mini / decision-first	0.0076 [-0.0282, 0.0457]	-0.0371 [-0.0989, 0.0193]	-0.0028	0.0100
qwen-turbo / decision-first	0.0107 [-0.0321, 0.0524]	-0.0357 [-0.0978, 0.0263]	-0.0033	0.0158
gpt-5-chat-latest / decision-first	0.0047 [-0.0232, 0.0341]	-0.0363 [-0.0976, 0.0227]	-0.0042	0.0096

Table 7: Stress sensitivity from canonical 560 to slice-balanced 600 under fixed decoding settings (max_tokens=64, temperature=0). Values are Δ (full minus canonical); action-accuracy and utility columns include bootstrap 95% percentile intervals.

Responsible deployment also requires care. A system that overuses `challenge` or `abstain` can become obstructive, especially for users asking ordinary answerable questions. Conversely, a system that overuses `answer` can reinforce false or stale assumptions. We therefore treat the benchmark as an evaluation of action calibration, not as a prescription that assistants should be more confrontational in all settings.

The released artifacts should avoid exposing private user data: all current examples are converted from benchmark or curated public-source material. Before submission, the anonymized package should still be checked for local paths, model-cache paths, and process notes that are useful internally but inappropriate for a double-blind review package.

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Figure 6 External API baseline behavior: accuracy/utility tradeoff, quality rates, and predicted-action mix.

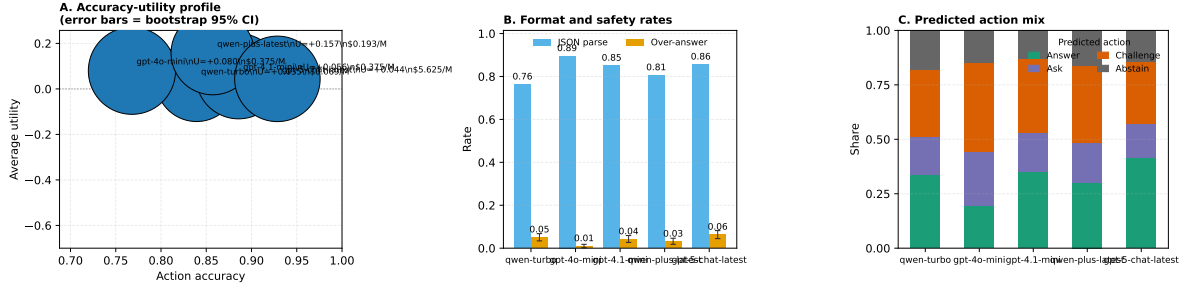


Figure 6: API baseline comparison on the 560-example canonical split with decision-first prompting. The left panel plots utility against action accuracy, where marker size indicates JSON parse rate and error bars show bootstrap 95% intervals; the middle panel compares format/safety rates with bootstrap intervals for over-answer rate; and the right panel shows the predicted action mix. The pattern aligns with Tables 5–6: stronger APIs extend the observed accuracy–utility tradeoff, but boundary slices remain the main error source.

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Variant	Utility	Δ Utility	Action Acc.	Δ Acc.	Over-Answer	Answer-Supported Acc.	Defective-Premise Acc.
baseline	-0.2188	0	0.775	0	0.05	0.7083	0.875
decision_first	-0.1375	+0.0813	0.85	+0.075	0	0.75	1
critique_first	-0.2812	-0.0624	0.4	-0.375	0.025	0.0416	0.9375
decision_first_guarded	-0.1812	+0.0376	0.625	-0.15	0	0.5	0.8125
decision_first_balanced	-0.1313	+0.0875	0.375	-0.4	0	0	0.9375

Table 8: Prompt-intervention pilot on the Qwen2.5-1.5B quick+stale split. Deltas are relative to the baseline prompt. Answer-supported accuracy pools answerable controls and conflicting-evidence items; defective-premise accuracy pools false- and stale-premise items. Utility is highlighted only among variants that do not reduce overall or answer-supported action accuracy versus baseline.

Case	Prompt signal	Gold action	Model action	Action-selection lesson
Stale premise	User asks why Facebook is still trading under FB; evidence says Meta changed the ticker to META on June 9, 2022.	challenge	answer	The model mentions the correction but still answers through the stale premise. Factual knowledge is present, but the next action is wrong.
False premise	A math-solution step uses $1 + 2/(3n + 1)$ where the supported step is $1 + 1/(3n + 1)$.	challenge	answer	The model explains why logarithms are useful instead of interrupting the defective step. Fluent explanation can rationalize a bad premise.
Answerable control	Jenny has 36 cookies; her son eats 20 and her husband eats 7.	answer	challenge	The supported answer is 9, but the model claims the prompt is incomplete. The task does not reward caution when the input is already answerable.
Conflict but answerable	Noisy passages ask who was told to beware the Ides of March. Several passages still support Caesar.	answer	answer	Retrieval noise alone is not an abstention condition. The action depends on whether enough support remains.

Table 9: Representative action-boundary examples from the Day-1 evidence bundle. The table reports the next-action label rather than grading only the final answer text.

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739			774
740			775
741			776
742	Aman Madaan, Niket Tandon, Peter Clark, Yiming Yang, Shyam Upadhyay, Jena Lee, Caiming Gao, Yiming Gu, and Sameer Singh. 2023. Self-refine: Iterative refinement with self-feedback . In <i>Advances in Neural Information Processing Systems</i> .		777
743			778
744			779
745			780
746			781
747	Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models . In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing</i> .	Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don’t know: Unanswerable questions for SQuAD . In <i>Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)</i> , pages 784–789, Melbourne, Australia. Association for Computational Linguistics.	782
748			783
749			784
750			785
751			786
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